Open-Source Continuous-Culture Bioreactors: A Correction, a Comparison, and a Two-Instrument Proposal

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Prepared with AI assistance (Claude, Anthropic).
Source repository: https://github.com/m9h/biopunk-bioreactor

10 July 2026 • Working paper — a reader's audit, not peer review

Abstract

Low-cost open-source continuous-culture devices — turbidostats, chemostats, and morbidostats — have proliferated, but they are frequently compared as though they compete for a single role. They do not. We argue that the field serves *three* distinct purposes: in-situ *characterisation* of a culture held in a defined state; automated *evolution* that must produce archived, replicated evidence; and turnkey *scale-out* operation. We first correct a set of verifiable errors and overstatements in the recent OpenEvo preprint [2], including a self-contradictory growth-rate figure and a false claim about a competitor's licensing. We then place six platforms — Chi.Bio, OpenEvo, Pioreactor, eVOLVER, EVE, and the Toprak morbidostat — in a like-for-like matrix across fifteen attributes, and give per-platform strengths and limitations. Finally we propose that the best system is not one merged machine but *two* purpose-built instruments sharing a component ecosystem, and we note the strong educational role these devices already play. Every factual claim is tagged with its confidence level; specifications not verified against a primary source are flagged as such.

1 Introduction

The precision of an in-vivo biology experiment depends on isolating the subsystem of interest from the drift of a growing culture's environment. Continuous-culture devices address this by diluting a culture during growth: a turbidostat holds optical density (OD) at a set-point; a chemostat fixes the dilution rate and lets density float; a morbidostat titrates a drug to hold growth inhibition constant. Over the past decade a wave of open-source, low-cost instruments has made these techniques broadly accessible, and two in particular frame the present analysis: **Chi.Bio** [1], an all-in-one platform for in-situ measurement and manipulation, and **OpenEvo** [2], a recent preprint describing a cheap, buildable device for automated directed evolution.

These two are often read as competitors — both cost roughly $300, both are open — but they are optimised for opposite halves of the problem. Chi.Bio is fundamentally an *instrument*: it exists to measure a culture richly while holding it in a defined state. OpenEvo is fundamentally an *actuator*: it exists to push a culture through selection regimes, unattended, for weeks. Recognising this distinction — and a third purpose that a productised platform such as Pioreactor serves — reorganises the entire field and changes what "best" means.

This working paper has four aims. Section 2 sets out the three purposes. Section 3 corrects specific errors in the OpenEvo preprint, fairly and with proposed fixes. Section 4 compares six platforms attribute by attribute, and Section 5 argues for building two purpose-built instruments rather than one compromise machine. Section 6 documents the educational materials these platforms already serve. We are explicit throughout about what was verified against a primary source and what was not.

2 Three purposes, not one

Treating all continuous-culture devices as interchangeable obscures that they optimise for incompatible objectives.

`proto` — **Characterisation.** Hold a culture in a defined, unchanging state and measure it richly, in situ, in real time, closing feedback loops around it. The figure of merit is *signal density per reactor*, with a handful of reactors running different conditions in parallel. Chi.Bio is the exemplar: a chip spectrometer, a seven-colour LED for optogenetic actuation and fluorophore excitation, non-contact infrared thermometry, and proportional-integral-derivative plus model-predictive control that holds OD within about 2% of set-point [1].

`devo` — **Evolution.** Push a culture through selection regimes unattended for weeks, and produce *evidence* that adaptation occurred and which mutation caused it. The figures of merit are cheap chambers in numbers, systematic archiving, and long-run robustness. This is OpenEvo's target, and — as Section 3 argues — the half it does not yet fully reach.

`bior` — **Scale-out operation.** Run many reliable reactors now, with minimal build effort, a maintained software platform, cluster management, and support. The value is reproducibility and uptime, not measurement richness or archived evidence. Pioreactor (turnkey) and eVOLVER (maximally configurable) own this axis, which is orthogonal to the other two.

These purposes pull hardware in opposite directions — rich-and-few versus cheap-and-many, non-contact versus immersed-electrode-tolerant, defined-steady-state versus free-to-run-to-stationary. A device optimised for one is measurably worse at another. That tension is the organising principle of everything below.

3 Corrections to the OpenEvo preprint

OpenEvo is a genuine contribution and the corrections here are not a dismissal. Its real strengths include a peak-to-peak growth readout tied to hardware dilution events (which sidesteps local-maxima noise); a 940 nm OD path that avoids optogenetic and ambient overlap, with an inverse model holding $R^2 \geq 0.98$ out to $OD_{600} = 6.0$; a closed autoclavable vessel with better sterility than Chi.Bio's open tube; a 4,000-hour remote run demonstrating robustness; and an illustrated assembly manual validated by having an inexperienced undergraduate build a working unit. It is the cheapest buildable three-media selection cycler in the field. The issues below are correctable and concern framing and analysis more than hardware.

3.1 Verifiable errors

[error] **The headline growth-rate figure contradicts itself.** The abstract states evolved cells "grew 55% faster than wild-type at 12% salt"; the results state "a doubling time of 4.9 hours versus 7.6 hours (a 36% increase in growth rate)." Both describe the same data, but the two numbers are different quantities. A doubling time of $7.6 \rightarrow 4.9$ h is a 36% *reduction in doubling time* yet a 55% *increase in growth rate*, since rate $\propto 1/(\text{doubling time})$ and $(1/4.9)/(1/7.6) = 1.55$. The abstract's 55% is the correct growth-rate figure; the results sentence mislabels the

36% doubling-time reduction as a growth-rate increase. *Fix:* state both explicitly — "36% shorter doubling time, equivalently 55% higher growth rate."

[error] **A false claim about a competitor.** The preprint states that Pioreactor is "available for purchase, but only the software is open source." A public `Pioreactor/hardware` repository exists and the hardware is openly licensed under CC BY-SA 4.0 (per its `LICENSE` file) [6]. The claim is used to clear competitive space and is false. *Fix:* describe Pioreactor accurately as open-hardware-and-software, distinguished not by openness but by being a supported product with a fleet platform. (We note in passing that an earlier draft of this document itself misnamed the licence as CERN-OHL; a contributor corrected it against the primary `LICENSE` file — an illustration of the citation discipline this repository aims for.)

3.2 Framing and omission

[framing] **The nearest prior art is uncited.** Chi.Bio [1] is open hardware and software, roughly the same $300, performs in-situ measurement and continuous culture, and predates OpenEvo by six years. The preprint surveys eVOLVER, the Toprak morbidostat, and Pioreactor, then frames an unfilled gap while omitting the system that most directly occupies it, which makes the gap look larger than it is.

[framing] **An unsupported superlative.** "The most affordable modern continuous growth system available" is not established: Pioreactor ships turnkey at $329 with no build, and EVE builds a triplicate system for $115–200 [4]. *Fix:* qualify to "among the lowest-cost *buildable* systems with three-media selection."

[framing] **The success is overstated.** The 9% salt step did not partially adapt — growth ceased, OD flat-lined, and the run terminated at cycle 152. The adaptation claim rests on a single hand-picked glycerol stock at cycle 145, assayed offline. The result is real, but the framing should foreground one successful selection step with a failed extension, not a ladder.

3.3 Methodological limitations

[method] $n = 1$ **forecloses causal attribution.** A single clone from a single population was sequenced. Without independent replicate lineages there is no convergence signal, and the reported single-nucleotide polymorphisms and two large deletions cannot be attributed to the low-salt adaptation. This is the field's central inference requirement — the Long-Term Evolution Experiment runs twelve populations for exactly this reason [8] — and it is the most consequential limitation.

[method] **The wrong variant caller.** Structural variants were called with "Geneious's default mapper with low-medium sensitivity for 5 iterations." The key finding — a ∼110 kb pHV4 deletion with "primarily transposon-associated elements" — is IS-mediated rearrangement, precisely the case a default short-read mapper handles worst. `breseq` exists for microbial-evolution resequencing including IS mobilisation [10], and its author is cited five times in the preprint. *Fix:* re-call with `breseq` and confirm the pHV4 structure with long reads.

[method] **No systematic archiving.** The preprint cites Barrick et al. — ". . . Archiving Populations, and Measuring Fitness. . . " [7] — as motivation, then implements neither archiving nor fitness measurement in the automated system. Head-to-head ancestor-versus-descendant competition, the gold standard the cited work describes, is impossible without a time-series archive.

Table 1: Six open continuous-culture platforms, attribute by attribute. Green marks a notable strength, red a notable weakness, amber a qualified case.

Attribute	**Chi.Bio**	**OpenEvo**	**Pioreactor**	**eVOLVER**	**EVE**
Primary purpose	proto	devo	bior	bior	devo
Reference	PLoS Biol '20	bioRxiv '26	none	Nat Biot '18	eLife '22
Parallelism	8 / computer	1 chamber	cluster	16 vials	≥3 reps
Working vol.	12–25 mL	∼20 mL	15–30 mL	∼40 mL	∼12 mL
OD source	650 nm laser	940 nm IR	900 nm IR	LED/PD	LED/PD
Rich sensing	spectrometer	OD only	plugin	config.	OD only
Contact	non-contact	closed	contact	contact	contact
Sterility	open air	closed, autocl.	autocl. vial	autocl. vial	3D-print
Selection	2+2 pumps	3 media+waste	pumps	config.	drug line
Control law	PID + MPC	threshold	Kalman GR	config.	threshold
Archiving	none	manual 1×	none	none	none
Open HW/SW	both	both	both	both	both
Cost/unit	$300+PCB	∼$300	$329–389	<$2k/16	$115–200
Build effort	PCB fab	guided	turnkey	machine shop	moderate
Independent use	ReacSight	too new	community	ePACE	teaching

4 A like-for-like comparison

Table 1 places six platforms across fifteen attributes. Purpose codes are proto (characterise), devo (evolve), bior (scale-out). Cells reflect each platform as published or shipped, not what a motivated builder could add; "independent use" means peer-reviewed work by groups other than the originators.

The Toprak morbidostat [5] is omitted from the table for width but belongs to purpose devo: it is the canonical antibiotic-resistance device (working volume ∼12 mL), titrating a drug to constant inhibition, widely adapted but a dated bespoke build rather than a maintained platform.

Two patterns are worth naming. First, *every platform lacks archiving* — the single most important capability for turning an evolution run into evidence, and a clear gap for the whole field. Second, the only platform without a reference paper is Pioreactor, which is also the only pure bior product; its credibility is commercial and community rather than bibliographic, a genuine consideration when an instrument must be cited.

5 A two-instrument proposal

Because the three purposes pull hardware in opposite directions, the strongest system is not one merged machine but two purpose-built instruments that share a component ecosystem.

5.1 The best characterisation instrument

Start from Chi.Bio; its architecture is already correct. The governing constraint is that *non-contact measurement is inviolable* — every subsystem but the heat plate reads through the vessel wall, which is why tubes hot-swap and sterilisation is trivial. Upgrades that respect this: dissolved-oxygen and pH optode spots read optically through the wall; a dispersive micro-spectrometer replacing eight fixed filter bands with a continuous 340–850 nm trace; and a sealed, gas-controlled lid to close Chi.Bio's open-to-atmosphere gap. Viable-cell-density by dielectric spectroscopy is explicitly *excluded* here because it requires immersed electrodes — it belongs on the other machine.

5.2 The best evolution machine

Start from OpenEvo; its virtues (cheap, closed, buildable, standalone) fit long unattended runs. But its two missing capabilities are not sensors. First, *replicate populations*: $n = 1$ is a failure of causal inference, so the build is eight cheap chambers under one coordinating interface, not one richly instrumented reactor. Second, *archiving*: an automated chilled fraction collector on the existing waste line, diverting a programmable fraction on a generation-count schedule into a chilled glycerol archive — the capability that makes fitness measurable, and, as far as we can determine, one nobody has built. A *batch-excursion* firmware mode — periodically suspending dilution to run to stationary phase — would additionally recover carrying capacity, a phenotype a turbidostat structurally cannot observe, and the very axis on which the OpenEvo cycle-145 population differed. The adaptive-laboratory-evolution literature adds a crucial caution: selection is for growth, and product formation costs growth, so growth-coupled strain design must precede selection [9].

5.3 Why two, not one

The sensor modalities sort themselves by contact. Optode DO/pH and dispersive spectrometry are optical and non-contact, and land on the characterisation instrument; conductivity, electrical resistance tomography, and dielectric viable-cell-density need immersed electrodes and land on the evolution machine, whose closed autoclavable vessel welcomes them. Almost no sensor is wanted by both and available to only one — the clearest evidence these are two instruments. What they *share* is an ecosystem: a common module footprint, reusable sensor-module designs, one firmware framework, and a shared calibration reference. Combine at the platform level, not the product level.

6 Educational role

These platforms are teaching instruments as much as research tools, and because one device spans microbiology, electronics, firmware, control theory, and data analysis, a single unit serves biology, engineering, and computing classes at once. EVE is the education-first exemplar: its paper is explicitly titled "…and its educational use," it builds for under $200, and it was validated in a real classroom evolving replicate *E. coli* under chloramphenicol [4]. Pioreactor maintains an educators portal and lesson concepts spanning OD-by-scattering through evolution. OpenEvo's assembly manual doubles as a build-it-yourself course and anchored an international hackathon in which students remotely controlled a device on another continent. Chi.Bio's own documentation offers its core features as a stand-alone teaching platform on which students can observe fluorescence by eye. Notably, the teaching materials inherit the research blind spots: archiving, replicate-based inference, and carrying capacity are as under-taught as they are under-implemented — an opportunity for a device-agnostic curriculum.

7 Discussion and limitations

This is a reader's audit, not peer review, and confidence is uneven. Full texts of Chi.Bio and OpenEvo were read directly. Pioreactor specifications, pricing, hardware licensing (CC BY-SA 4.0, verified against the primary `LICENSE` file), and the absence of a reference paper were confirmed from primary or product sources, as were the headline specifications of eVOLVER and EVE and the working volume of the Toprak morbidostat. Component datasheets underpinning the proposed evolution machine (spectrometer, spectral sensor, and impedance front-ends) were verified in a companion pass. Some adoption specifics and a small number of vendor register-level details remain unverified at page level and are flagged as such in the repository's `SOURCES.md`. No claim here should be budgeted against without confirmation at its primary source.

8 Conclusion

The open-bioreactor field is best understood as serving three purposes, not one contest. Chi.Bio owns characterisation; Pioreactor and eVOLVER own scale-out; and evolution — the purpose with the most devices chasing it — has no clear champion, because no platform yet archives or replicates as the science requires. OpenEvo's real contribution is the cheapest buildable selection cycler with a genuinely excellent manual; its preprint's errors of framing and analysis are correctable and worth correcting. The best path forward is two purpose-built instruments sharing a parts ecosystem, with archiving and replication treated as first-class — and with the educational mission, already strong, extended by a curriculum that teaches the methodological rigour the hardware has so far left out.

Source data, the full comparison matrix, per-platform pros and cons, a datasheet-verification pass, an educational-materials guide, and this document's LaTeX sources are maintained at `https://github.com/m9h/biopunk-bioreactor` *under CC BY 4.0. Corrections with citations are welcome.*

References

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